```
*** EXAMPLE CONSOLE INTERACTION FROM MAIN.PY ***


A: Integrity Check
B: Convert text  ---> LaTex
C: Convert Latex ---> text
D: Update latex
E: Select exercises from exercise bank
F: Create solution file
G: Create new problem set with same kinds of problems
X: Exit
Option: A
File to check (hit Enter to default to exercise manual):
Number of exercises: 1688
Good - No duplicate exercises!


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Option: E
File with database spec (hit Enter to default to exercise manual):
For category ' type ' possible values: ['LECTURE', 'CLASS', 'HW', 'TEST',
'GROUP']
Your choice (hit enter for NULL):TEST

For category ' diff ' possible values: ['EASY', 'MEDIUM', 'HARD', 'CREATIVE']
Your choice (hit enter for NULL):

For topics options: ['ALG', 'CRYPTO', 'DEF', 'DISC', 'DOM', 'EVAL', 'EXP',
'FRAC', 'FUNCT',
'GRAPH', 'GRAPHTHEORY', 'INDUCTION', 'INT', 'INVERSE', 'LOG', 'MATRIX',
'MODARITH', 'NUM',
'NUMBASE', 'NUMTHY', 'PROG', 'PROOF', 'PROPLOGIC', 'PYTHON', 'QUANT',
'RANGE', 'RAD',
'RECURSIVE', 'RELATION', 'SET', 'STAT', 'TREE', 'TRIG', 'WORD']
Your choices space or comma separated (hit enter for none):TREE

Path to template file (hit enter for default):
..\Templates\DatabaseResults.txt
Output file name (hit enter for default name):
Database results at: OutputFilesDefault\DataBaseResults.txt


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Option: B
Path to file, excluding file name (hit enter for 'InputFilesDefault'):
File name:DataBaseResults.txt
Name of EXERCISE file (hit enter for default manual):
Output file path, excluding file (hit enter for 'OutputFilesDefault'):
Output TEX file at: DataBaseResults.tex
In folder:  ..\OutputFilesDefault\
```

```
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X: Exit
Option: D
Path to file, excluding file name (hit enter for 'InputFilesDefault'):
File name:DataBaseResults.tex
Name of EXERCISE file (hit enter for default manual):
Output file path, excluding file (hit enter for 'OutputFilesDefault'):
Update Latex in: ..\OutputFilesDefault\
```

Updates LaTex

← Manually moved LaTex here

↑ Updated LaTex written to default location

```
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E: Select exercises from exercise bank
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X: Exit
Option: F
Directory of file (hit enter for default of 'InputFilesDefault'):
File name:DataBaseResults.tex
OpenAI model to use (hit enter for default `gpt-6-luna'):
Your secret OpenAI API key: [PUT YOUR OPEN AI KEY HERE]
Directory where you want output file (hit enter for default of
'OutputFilesDefault'):
Complete solutions written to SOLUTIONS_DataBaseResults.tex
In folder:  ..\OutputFilesDefault\
```

New LaTex with solutions added

↓ Same file, but with solutions filled in written to default location

```
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D: Update latex
E: Select exercises from exercise bank
F: Create solution file
G: Create new problem set with same kinds of problems
X: Exit
Option: G
Directory of file (hit enter for default of 'InputFilesDefault'):
File name:DataBaseResults.tex
OpenAI model to use (hit enter for default `gpt-6-luna'):
Your secret OpenAI API key: [PUT YOUR OPEN AI KEY HERE]
Create solutions too (Y/N):N
For new codes in the problem, we need some info...
First, we need today's date.
Enter year: 2026
Enter month: 10
Enter day: 23
Number for the first exercise (hit enter for 1):
Directory for new file (hit enter for default of 'OutputFilesDefault'):
New file written to NEW_DataBaseResults.tex
```

Creates doppelganger

↳ New problem set written to default location

```
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```
D: Update latex
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F: Create solution file
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X: Exit
Option: X
```

↳ Exit